



Khayal

Medication Reminder App

Software Requirements Specification (SRS)

Version 1.1 — Updated & Complete

May 2026

For internal team use — Designers · Developers · Evaluators

1. Introduction

This document explains what the Khayal app does and how it should work. It is written simply so everyone on the team — designers, developers, and evaluators — can understand it easily.

1.1 What is This Document For?

This SRS (Software Requirements Specification) is a detailed guide that answers three questions:

- What problem does Khayal solve?
- Who will use the app?
- What exactly must the app do?

1.2 What Does Khayal Do?

Khayal is a mobile app that helps elderly Pakistani people (aged 60+) remember to take their medicines on time. It also helps their family members keep track and get notified if a medicine is missed.

The app works in two languages:

- Urdu (main language — text goes right to left)
- English (second language — text goes left to right)

What is INCLUDED in the first version (MVP):

- Medicine reminders
- Family alert system
- Simple home screen for elderly users
- Dashboard for family caregivers
- Language and font size settings

What is NOT included yet (for future versions):

- Doctor portal
- WhatsApp bot
- Automatic pill identification using camera

1.3 Key Words You Should Know

Word	Simple Meaning
Elder User	The elderly person (age 60+) who takes the medicine
Caregiver	A family member (son, daughter, spouse) who manages reminders

Word	Simple Meaning
Dose Confirmation	When the elderly user taps 'Taken' to confirm they took medicine
Escalation	When the app sends a louder alert or notifies family after no response
RTL	Right-to-Left — how Urdu text is written (opposite of English)
SUS Score	A score (out of 100) that measures how easy the app is to use
MVP	Minimum Viable Product — the basic version we build first
Offline Mode	When the app works without an internet connection
Local Notification	An alarm stored on the phone itself — fires without internet
Push Notification	A message sent from the server to the phone — requires internet
Encryption	Scrambling data so it cannot be read by unauthorized people
OTP	One-Time Password — a 6-digit code used to link two accounts

2. The Problem and Our Solution

2.1 The Problem

Many elderly people in Pakistan forget to take their medicines. Here is why:

- They take many medicines (2 to 5 different ones daily)
- They may not read well or understand complex apps
- Existing apps are in English and are hard to use
- Family members have no way to track if medicine was taken

2.2 Khayal's Solution

Khayal solves this with three key features:

- Simple reminders — a loud voice reminder in Urdu at the right time
- One-tap confirmation — the elderly user just presses ONE big button to confirm
- Family alerts — if medicine is missed, family is notified automatically

2.3 How the System Works (Big Picture)

Part	Who Uses It	What It Does
Elder View	Elderly user	Shows today's medicines in large Urdu text. Sends reminders. Has one big button.
Caregiver Dashboard	Family member	Shows medicine schedule, missed doses, and weekly charts. Sends alerts.
Notification Engine	Runs automatically	Fires reminders on time, escalates if no response, alerts family after 30 min.

3. Who Uses the App?

There are three types of users:

3.1 Primary User: The Elderly Patient

Detail	Description
Age	60 to 85 years old
Location	Urban/semi-urban Pakistan (Lahore, Karachi, Islamabad)
Tech Skills	Basic — can use WhatsApp but not complex apps
Health	Usually has 2–5 ongoing health conditions
Language	Prefers Urdu or regional languages
Special Needs	Needs large text, voice prompts, and very few buttons

3.2 Secondary User: The Family Caregiver

Detail	Description
Age	25 to 55 years old
Relationship	Son, daughter, daughter-in-law, or spouse
Location	Same house OR far away (another city or abroad)
Tech Skills	Good — comfortable with smartphones
Main Concern	Knowing in real time whether their parent/spouse took medicine

3.3 Tertiary User: The Doctor (Future)

In a future version, the doctor will be able to upload prescriptions and see adherence reports. This is not part of the first version.

4. What the App Must Do (Functional Requirements)

This section lists everything the app must do. Each requirement has an ID, a simple description, and a priority (Must Have = required, Should Have = important but not critical).

4.1 Step 1: Opening the App for the First Time

When someone opens the app for the first time, these things must happen in order:

ID	What Must Happen	Priority
FR-AUTH-01	The app asks: 'Are you a Patient or a Caregiver?' — user picks one	Must Have
FR-AUTH-02	The app shows a language chooser (Urdu or English) before anything else	Must Have
FR-AUTH-03	The caregiver enters a 6-digit OTP code to link their account to the patient's phone	Must Have
FR-AUTH-04	The caregiver can set up everything for the patient — patient does not need to touch settings	Must Have
FR-AUTH-05	Once logged in, the app remembers the user — they do not need to log in again	Must Have

4.2 Step 2: Adding Medicines

After setup, the caregiver adds medicines for the patient:

ID	What Must Happen	Priority
FR-MED-01	Caregiver can add a medicine with: name (in Urdu and English), dose amount, type (tablet/capsule/ml), how many times a day, and what time(s) to take it	Must Have
FR-MED-02	Caregiver can add a photo of the pill so the elder can recognize it	Should Have
FR-MED-03	The patient's home screen shows medicines in time order (earliest first)	Must Have
FR-MED-04	Caregiver can edit or delete any medicine at any time	Must Have
FR-MED-05	Medicine name shows in Urdu (big) and English (small, below)	Must Have

4.3 Step 3: Sending Reminders

At the scheduled time, the app automatically alerts the elderly user:

ID	What Must Happen	Priority
FR-REM-01	A notification appears on the phone at the scheduled time	Must Have
FR-REM-02	The notification shows the medicine name (Urdu), dose, and two buttons: 'Taken' and 'Remind me in 15 min'	Must Have
FR-REM-03	The phone plays a sound when the reminder fires (user can choose the sound)	Must Have
FR-REM-04	The phone also vibrates with the reminder	Must Have
FR-REM-05	The reminder works even if the app is closed or running in the background	Must Have
FR-REM-06	If the user taps the notification, the app opens showing a big 'Mark as Taken' button	Must Have
FR-REM-07	If the user taps 'Snooze', the reminder fires again exactly 15 minutes later	Must Have

4.4 Step 4: What Happens If Medicine Is Missed

If the elderly user does not respond to the reminder, the app automatically escalates:

ID	What Must Happen	Priority
FR-ESC-01	After 15 minutes of no response → app sends another louder reminder	Must Have
FR-ESC-02	After 30 minutes of no response → app sends an alert to the family caregiver's phone	Must Have
FR-ESC-03	The caregiver alert says: patient name, medicine name, scheduled time, and how long ago the reminder was sent	Must Have
FR-ESC-04	Once the elderly user taps 'Taken', all alerts stop automatically	Must Have
FR-ESC-05	Every escalation event is saved with the exact date and time	Should Have

4.5 Step 5: Recording What Was Taken

ID	What Must Happen	Priority
FR-LOG-01	Every dose confirmation is saved with: medicine name, exact time, and status (Taken / Missed / Snoozed)	Must Have
FR-LOG-02	On the home screen, each medicine shows one of three labels: Taken ✓ / Upcoming ⌚ / Missed ✗	Must Have

ID	What Must Happen	Priority
FR-LOG-03	The app saves at least 90 days of history	Must Have
FR-LOG-04	Patient can see a simple 7-day history view	Should Have

4.6 Caregiver Dashboard

ID	What Must Happen	Priority
FR-CARE-01	Dashboard shows today's status for each medicine (Taken / Missed / Upcoming) in real time	Must Have
FR-CARE-02	Dashboard shows a weekly bar chart — how many medicines were taken each day (Mon–Sun)	Must Have
FR-CARE-03	Caregiver gets a push notification on their phone every time a dose is missed	Must Have
FR-CARE-04	Caregiver can add, edit, or delete medicines directly from the dashboard	Must Have
FR-CARE-05	Dashboard shows total number of medicines and the overall adherence percentage	Should Have

4.7 Language and Accessibility

ID	What Must Happen	Priority
FR-L10N-01	All text in the app is available in both Urdu and English	Must Have
FR-L10N-02	When Urdu is selected, the entire screen layout flips to right-to-left	Must Have
FR-L10N-03	Urdu text uses the Noto Nastaliq Urdu font (clear and readable)	Must Have
FR-L10N-04	User can choose from 3 font sizes: Normal (18sp), Large (22sp), Extra Large (26sp)	Must Have
FR-L10N-05	The app speaks the medicine name out loud in Urdu when a reminder fires	Should Have
FR-L10N-06	All buttons are at least 48×48dp in size — big enough for elderly fingers	Must Have
FR-L10N-07	Text and background colors must have enough contrast (at least 4.5:1 ratio)	Must Have

5. Performance and Reliability Requirements

These are rules about HOW WELL the app must work, not just what it does.

5.1 Offline Reliability (NEW)

This is a critical requirement for the Khayal app because elderly users in Pakistan frequently turn off Wi-Fi or mobile data to save battery. The app must distinguish between what requires the internet and what does not.

Why This Matters

- If reminders only fire when there is internet, the app fails its core purpose whenever the user loses connectivity.
- The family alert system correctly requires internet — the app must communicate this gracefully to caregivers.

ID	What Must Happen	Priority
NFR-OFF-01	All medicine reminders (FR-REM-01 through FR-REM-07) must work without an internet connection. They use locally stored alarms on the device.	Must Have
NFR-OFF-02	The 'Mark as Taken' button and dose logging (FR-LOG-01, FR-LOG-02) must work without internet. The log is saved locally and synced to the server when connectivity returns.	Must Have
NFR-OFF-03	The caregiver push notification (FR-ESC-02, FR-CARE-03) requires internet. If there is no internet, the alert must be queued and sent the moment connectivity is restored.	Must Have
NFR-OFF-04	The caregiver dashboard (FR-CARE-01, FR-CARE-02) requires internet to show real-time data. If offline, the app must display the last-synced data with a visible 'Last updated: [time]' label.	Must Have
NFR-OFF-05	The patient's home screen (medicine list, upcoming doses) must load from locally cached data and display correctly even with no internet.	Must Have

5.2 Security and Privacy (NEW)

Because Khayal handles personal health schedules and family contact information, it must protect user data properly. This is especially important as the app links two people's accounts together.

Why This Matters

- Medication schedules reveal private health information. They must not be readable by unauthorized parties.
- The OTP linking system (FR-AUTH-03) must be tamper-proof to prevent a stranger from linking to a patient's account.
- Phone numbers and names must not be exposed or sent unencrypted.

ID	What Must Happen	Priority
NFR-SEC-01	All data sent between the app and the server must be encrypted using HTTPS/TLS 1.2 or higher. Unencrypted HTTP connections are not permitted.	Must Have
NFR-SEC-02	All user data stored in the database (medicine names, schedules, phone numbers, logs) must be encrypted at rest.	Must Have
NFR-SEC-03	The 6-digit OTP used to link accounts (FR-AUTH-03) must expire after 10 minutes and become invalid after one successful use.	Must Have
NFR-SEC-04	The app must not store passwords in plain text. Any authentication tokens must be stored securely using the device's secure storage (Android Keystore / iOS Keychain).	Must Have
NFR-SEC-05	The app must not share or sell user data to any third party. This must be stated clearly in a Privacy Policy accessible from the Settings screen.	Must Have
NFR-SEC-06	A caregiver can only view data for patients they are explicitly linked to. One caregiver cannot see another patient's data.	Must Have
NFR-SEC-07	Failed OTP attempts must be rate-limited: after 5 wrong attempts, the code is invalidated and a new one must be requested.	Should Have

5.3 Device Permissions (NEW)

To deliver reminders reliably, the app needs specific permissions from the phone's operating system. These must be requested clearly and explained in simple language to the user.

Why This Matters

- Without 'Do Not Disturb' bypass permission, a reminder will be silenced if the phone is on silent mode — exactly when an elderly user is asleep and most likely to miss a dose.
- Without 'Draw over other apps' or lock-screen permission, the big 'Mark as Taken' button cannot appear when the phone is locked.
- Without exact alarm permission (Android 12+), scheduled reminders may be delayed by the OS to save battery.

Permission	Why It Is Needed	Which Platform	Priority
POST_NOTIFICATIONS	Required to show any notification on Android 13 and above	Android 13+	Must Have
USE_EXACT_ALARM / SCHEDULE_EXACT_ALARM	Ensures reminders fire at the precise scheduled time, not delayed by battery saving	Android 12+	Must Have
RECEIVE_BOOT_COMPLETED	Re-registers all alarms after the phone is restarted, so reminders continue working	Android	Must Have
Notification (UserNotifications)	Required to show alerts and sounds on iOS devices	iOS	Must Have
Critical Alerts (entitlement)	Allows sound to play even when iPhone is on Silent or Do Not Disturb mode	iOS	Must Have
SYSTEM_ALERT_WINDOW (Draw over apps)	Allows the big 'Mark as Taken' button to appear on top of the lock screen	Android	Should Have
VIBRATE	Enables phone vibration with each reminder (FR-REM-04)	Android	Must Have
CAMERA (future)	For pill photo feature (FR-MED-02). Must only be requested when the feature is used.	Both	Should Have

Important: All permissions must be requested at the moment they are first needed — not all at once during onboarding. The app must explain in simple Urdu/English why each permission is being asked for.

5.4 Usability Targets

ID	Rule	Target
NFR-USE-01	Easy enough to use — measured with a usability score	SUS score of 75 or above (out of 100)
NFR-USE-02	Elderly user can confirm a dose very quickly	In 2 taps or fewer
NFR-USE-03	Most people complete setup without getting stuck	80% or more complete it without help

6. Screens in the App

Here is a complete list of every screen in the app, who uses it, and how important it is (P0 = must have, P1 = important but can come later):

Screen ID	Screen Name	Urdu Name	Who Uses It	Priority
SCR-001	Splash Screen (opening logo)	اسپلش	Both	P0
SCR-002	Language Select	زبان	Both	P0
SCR-003	Role Select (Patient or Caregiver)	کردار	Both	P0
SCR-004	Caregiver Registration	اندراج	Caregiver	P0
SCR-005	Patient Profile Setup	مریض	Caregiver	P0
SCR-006	Add Medication	دوا شامل کریں	Caregiver	P0
SCR-007	OTP Link Screen (linking accounts)	ربط	Both	P0
SCR-008	Patient Home Screen	گھر	Patient	P0
SCR-009	Dose Confirmation Screen	تصدیق	Patient	P0
SCR-010	Medication Detail View	تفصیل	Patient	P1
SCR-011	Patient History (past 7–90 days)	تاریخ	Patient	P1
SCR-012	Caregiver Dashboard	ڈیش بورڈ	Caregiver	P0
SCR-013	Medication Management (full list)	انتظام	Caregiver	P0
SCR-014	Edit Medication	ترمیم	Caregiver	P0
SCR-015	Alert History	انتباہ	Caregiver	P1
SCR-016	Settings (language, font size)	ترتیبات	Both	P0
SCR-017	Notification (phone OS overlay)	اطلاع	Patient	P0

7. Data — What Gets Saved

The app saves four types of information:

7.1 User Data (Who is using the app)

- User ID — a unique code for each person
- Role — Patient or Caregiver
- Name and phone number
- Preferred language (Urdu or English)
- Font size preference
- Link to the other user (patient-caregiver connection)

7.2 Medication Data (What medicines to take)

- Medicine ID — a unique code
- Medicine name in Urdu AND English
- Dose amount and type (tablet, capsule, ml)
- How often (once, twice, three times, or custom)
- Exact time(s) to take it
- Optional: photo of the pill
- Notes from caregiver
- Whether the medicine is still active (on/off)

7.3 Dose Logs (What was actually taken)

- Which medicine
- Scheduled time vs actual confirmation time
- Status: Taken / Missed / Snoozed / Escalated
- Whether and when the family was alerted

7.4 Alert Data (Family notifications sent)

- Which caregiver was alerted
- Which patient and which medicine triggered the alert
- When the alert was sent
- Type: Missed Dose or 30-Minute Escalation
- Whether the caregiver has read the alert

8. Design Rules

These are the design principles the app must follow. They ensure the app is friendly and easy for elderly Pakistanis.

8.1 Big, Clear, and Simple

- Minimum body text size: 18sp. Headings: 24sp or bigger
- High contrast — dark text on light background (at least 4.5:1 contrast ratio)
- All buttons are at least 48×48dp — easy to tap with one finger
- No swipe gestures — everything done with simple taps

8.2 Urdu First

- The whole app is designed for Urdu first — English is secondary
- Urdu uses Noto Nastaliq Urdu font for beautiful, readable text
- All text goes right-to-left when Urdu is selected
- Numbers shown in Eastern Arabic format when culturally preferred

8.3 Keep It Simple — Show Only What Is Needed

- The patient sees only 3 screens: Home → Reminder → History
- The caregiver sees more options — but advanced settings are hidden by default
- Complex features are never shown on the main screen

8.4 Family-Friendly Alerts

- Alert messages to family are polite and non-alarming
- The tone of alerts is respectful (this is important in Pakistani family culture)
- Caregiver is treated as the main helper — not an afterthought

8.5 Warm and Friendly Look

- Warm colors — soft greens, earthy tones (no cold hospital blues)
- Friendly Urdu writing — no medical jargon
- Positive messages when medicines are taken on time (e.g., 'آج آپ نے تمام دوائیں لے لیں!', 'شباباش')

9. Revision History

Version	Date	Changes Made	Author
1.0	May 2026	Initial SRS document created	Project Team
1.1	May 2026	Added Section 5.1 (Offline Reliability), 5.2 (Security & Privacy), 5.3 (Device Permissions). Fixed missing section numbering. Added glossary terms: Offline Mode, Local vs Push Notification, Encryption, OTP.	Project Team